

Lecture 15: Splitting Fields and Normal Extensions

MAT205 Abstract Algebra II

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Part I: Splitting Fields

Definition: Splitting Field

Splits. $f \in F[x]$ of degree $n \geq 1$ *splits* in an extension E/F if it factors into linear factors in $E[x]$:

$$f(x) = c(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n), \quad c \in F^\times, \alpha_i \in E.$$

Definition. E/F is a *splitting field* for f over F if

- (i) f splits in E , and
- (ii) $E = F(\alpha_1, \dots, \alpha_n)$ — generated over F by the roots of f .

Condition (ii) forces minimality. So $\mathbb{C}$ is *not* a splitting field of $x^2 - 2$ over $\mathbb{Q}$: it splits there, but $\mathbb{C} \neq \mathbb{Q}(\sqrt{2})$.

Examples. $x^2 - 2$ over $\mathbb{Q}$: $\mathbb{Q}(\sqrt{2})$. $x^2 + 1$ over $\mathbb{R}$: $\mathbb{C}$. $x^2 + x + 1$ over $\mathbb{F}_2$: $\mathbb{F}_4$.

Existence of Splitting Fields

Convention. Fix an algebraic closure $\overline{F}$ of F . By the isomorphism extension theorem (applied to id_F), every algebraic extension of F embeds into $\overline{F}$, so we view all of them as subfields of $\overline{F}$.

Theorem. Every $f \in F[x]$ of degree $n \geq 1$ has a splitting field E over F inside $\overline{F}$, with $[E : F] \leq n!$.

Proof.

- $\overline{F}$ is algebraically closed, so f has roots $\alpha_1, \dots, \alpha_n \in \overline{F}$. Put $E = F(\alpha_1, \dots, \alpha_n)$ — a splitting field.
- Adjoin the roots one at a time: $[F(\alpha_1) : F] \leq n$, then $[F(\alpha_1, \alpha_2) : F(\alpha_1)] \leq n - 1$ (the minimal polynomial of α_2 divides $f/(x - \alpha_1)$), and so on.
- Multiplying, $[E : F] \leq n(n - 1) \cdots 1 = n!$. □

Without a closure in hand, the roots can instead be built by the Kronecker step $F \rightsquigarrow F[x]/(p)$, iterated — exactly how $\overline{F}$ itself is constructed.

Uniqueness of Splitting Fields

Inside the fixed $\overline{F}$, the splitting field of f is the one subfield generated by its roots, $E = F(\alpha_1, \dots, \alpha_n)$ — so “the” splitting field is well-defined. The only question is whether a different algebraic closure gives an isomorphic field.

Theorem (Fraleigh, Exercise 50.25). Any two splitting fields of f over F are isomorphic by an isomorphism fixing F .

Proof.

- Let $E \subseteq \overline{F}$ and $E' \subseteq \overline{F'}$ be splitting fields of f in two algebraic closures of F .
- The two closures are F -isomorphic via some $\lambda : \overline{F} \rightarrow \overline{F'}$ (isomorphism extension theorem).
- λ fixes F , so it permutes the roots of f ; hence $\lambda(E) = F(\lambda\alpha_1, \dots, \lambda\alpha_n) = E'$.
- So λ restricts to an isomorphism $E \rightarrow E'$. □

Examples of Splitting Fields

- $x^3 - 2$ **over** $\mathbb{Q}$. Roots $\sqrt[3]{2}$, $\omega\sqrt[3]{2}$, $\omega^2\sqrt[3]{2}$ with $\omega = e^{2\pi i/3}$. Splitting field $\mathbb{Q}(\sqrt[3]{2}, \omega)$:

$$[\mathbb{Q}(\sqrt[3]{2}, \omega) : \mathbb{Q}] = \underbrace{2}_{\omega} \cdot \underbrace{3}_{\sqrt[3]{2}} = 6 = 3!.$$

- $x^4 - 2$ **over** $\mathbb{Q}$. Roots $\pm\sqrt[4]{2}$, $\pm i\sqrt[4]{2}$. Splitting field $\mathbb{Q}(\sqrt[4]{2}, i)$, degree 8.
- $x^p - 1$ **over** $\mathbb{Q}$ (p **prime**). With $\zeta = e^{2\pi i/p}$ a primitive p -th root of unity, the splitting field is $\mathbb{Q}(\zeta)$ (cyclotomic field), of degree $p - 1$.
- $x^{p^n} - x$ **over** $\mathbb{F}_p$. Its p^n distinct roots form the field $\mathbb{F}_{p^n}$ — so $\mathbb{F}_{p^n}$ is the splitting field of $x^{p^n} - x$.

Part II: Normal Extensions

Definition: Normal Extension

Definition. An algebraic extension E/F is *normal* if every irreducible $p \in F[x]$ with at least one root in E splits completely in E . Equivalently: E contains all conjugates of each of its elements.

Lemma. Every degree-2 extension is normal. (The second root of a quadratic is $-b/a$ minus the first, so it lies in E too.)

Non-example. $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is *not* normal: $x^3 - 2$ has the root $\sqrt[3]{2}$ but its other roots are non-real, so they are not in $\mathbb{Q}(\sqrt[3]{2}) \subset \mathbb{R}$.

Terminology. Fraleigh's "finite normal extension" (Def 53.1) means a *separable* splitting field — our *Galois* extension. His name for the property above is just "splitting field over F " (Cor 50.6). We keep the modern split.

Characterization of Finite Normal Extensions

Theorem (Fraleigh 50.3 + Cor 50.6). For a finite extension E/F inside $\overline{F}$, the following are equivalent.

- (1) E/F is normal.
- (2) E is the splitting field over F of some $f \in F[x]$.
- (3) Every F -embedding $\tau : E \rightarrow \overline{F}$ has image $\tau(E) = E$ — i.e. every such τ is an automorphism of E .

Fraleigh states (3) as: every automorphism of $\overline{F}$ fixing F maps E onto itself. This is the operational form — a normal extension has no “outward” embeddings, only self-maps.

Characterization: Proof

(1) $\Rightarrow$ (2). Write $E = F(\alpha_1, \dots, \alpha_r)$ and set $f = \prod_i m_{\alpha_i, F}$. Each $m_{\alpha_i, F}$ has a root in E , so by normality splits in E ; thus E is the splitting field of f .

(2) $\Rightarrow$ (3). An F -embedding τ sends each root of f to a root of f (it fixes the coefficients), so it permutes the finite root set. Since E is generated by those roots, $\tau(E) = E$.

(3) $\Rightarrow$ (1). Let $p \in F[x]$ be irreducible with a root $\alpha \in E$, and $\beta \in \overline{F}$ any root of p . The conjugation isomorphism $F(\alpha) \rightarrow F(\beta)$ extends to an embedding $\tau : E \rightarrow \overline{F}$ with $\tau(\alpha) = \beta$. By (3), $\beta = \tau(\alpha) \in E$. So p splits in E . $\square$

Normal Closure

Theorem. Every finite $E = F(\alpha_1, \dots, \alpha_r)$ sits inside a smallest normal extension of F : the splitting field N of

$$f = m_{\alpha_1, F} m_{\alpha_2, F} \cdots m_{\alpha_r, F},$$

called the *normal closure* of E/F .

- N is normal (a splitting field) and contains E .
- Any normal extension containing E contains every conjugate of each α_i , hence contains N — so N is minimal.

Example. The normal closure of $\mathbb{Q}(\sqrt[3]{2})$ is $\mathbb{Q}(\sqrt[3]{2}, \omega)$, the splitting field of $x^3 - 2$ — degree 6.

Normality Is Not Transitive

Consider the tower

$$\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt[4]{2}).$$

- Each step has degree 2, hence is normal.
- But $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$ is *not* normal: $x^4 - 2$ has the root $\sqrt[4]{2}$ but its roots $\pm i \sqrt[4]{2}$ are non-real, so not in $\mathbb{Q}(\sqrt[4]{2}) \subset \mathbb{R}$.

So two stacked normal steps need not give a normal extension. (By contrast, algebraicity is transitive.) What is always true: if E/F is normal, then E/K is normal for every intermediate K .

The Index $\{E : F\}$ and the Bound $\{E : F\} \leq [E : F]$

Definition. $\{E : F\}$ is the number of F -embeddings $E \rightarrow \overline{F}$.

The chain.

$$|\mathrm{Aut}(E/F)| \leq \{E : F\} \leq [E : F].$$

- $|\mathrm{Aut}(E/F)| \leq \{E : F\}$: every F -automorphism of E is in particular an F -embedding $E \rightarrow \overline{F}$ (its image is E).
- $\{E : F\} \leq [E : F]$: for $E = F(\alpha)$, an F -embedding sends α to a root of m_α , and m_α has at most $\deg m_\alpha = [F(\alpha) : F]$ roots, so $\{F(\alpha) : F\} \leq [F(\alpha) : F]$. Index and degree are both multiplicative over a tower, so the bound multiplies up to $\{E : F\} \leq [E : F]$.

Index and Normality

Two equalities in the chain $|\text{Aut}(E/F)| \leq \{E : F\} \leq [E : F]$, two properties:

- $|\text{Aut}(E/F)| = \{E : F\} \iff E/F$ normal (Fraleigh Cor 50.7): the embeddings with image E are exactly the automorphisms, so the counts agree iff every embedding has image E — which is normality.
- $\{E : F\} = [E : F] \iff E/F$ separable: the bound $\{F(\alpha) : F\} \leq [F(\alpha) : F]$ is an equality iff m_α has *deg* m_α *distinct* roots, i.e. no repeated roots — the definition of α separable; multiplicativity lifts this to E/F .

Separable regime. Over a perfect field (char 0 or finite — most of this course), $\{E : F\} = [E : F]$ automatically, so $|\text{Aut}(E/F)| = [E : F] \iff E/F$ normal.

Galois Extensions = Normal + Separable

Theorem. For a finite extension E/F ,

$$|\operatorname{Aut}(E/F)| = [E : F] \iff E/F \text{ normal and separable.}$$

Both inequalities $|\operatorname{Aut}(E/F)| \leq \{E : F\} \leq [E : F]$ become equalities: normal closes the left, separable the right.

Definition. Such an E/F is a *Galois extension*; one writes $\operatorname{Gal}(E/F) := \operatorname{Aut}(E/F)$. (This is Fraleigh's "finite normal extension", Def 53.1.)

- $\mathbb{F}_{p^n}/\mathbb{F}_p$: Galois, $\operatorname{Gal} \cong \mathbb{Z}/n\mathbb{Z}$.
- $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$: Galois, $|\operatorname{Gal}| = 4$.
- $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$: not Galois — $|\operatorname{Aut}| = 1 < 3$, the deficit is non-normality.

Next: the Galois correspondence between subgroups of $\operatorname{Gal}(E/F)$ and intermediate fields.